

Indian Institute of Technology Dharwad

CS201L: Artificial Intelligence Laboratory

Seaborn Library : Functions

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Seaborn

Seaborn is a popular Python library for creating attractive and informative statistical graphics. Built on top of Matplotlib and integrated with pandas data structures, it provides a high-level interface that simplifies complex data visualization tasks.

Table 1: Commonly Used Seaborn Functions, Descriptions, and Parameters

Function	Description	Parameters
sns.set()	Sets the aesthetic parameters for seaborn plots.	<code>seaborn.set(*args, **kwargs)</code>
sns.lineplot()	Draws a line plot with confidence intervals.	<code>seaborn.lineplot(data=None, *, x=None, y=None, hue=None, size=None, style=None, units=None, weights=None, palette=None, hue_order=None, hue_norm=None, sizes=None, size_order=None, size_norm=None, dashes=True, markers=None, style_order=None, estimator='mean', errorbar=('ci', 95), n_boot=1000, seed=None, orient='x', sort=True, err_style='band', err_kws=None, **kwargs)</code>

Function	Description	Parameters
sns.scatterplot()	Draws a scatter plot.	seaborn.scatterplot(data=None, *, x=None, y=None, hue=None, size=None, style=None, palette=None, hue_order=None, hue_norm=None, sizes=None, size_order=None, size_norm=None, markers=True, style_order=None, legend='auto', ax=None, **kwargs)
sns.barplot()	Draws a bar plot with statistical aggregation.	seaborn.barplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, estimator='mean', errorbar=('ci', 95), n_boot=1000, seed=None, units=None, weights=None, orient=None, color=None, palette=None, saturation=0.75, fill=True, hue_norm=None, width=0.8, dodge='auto', gap=0, native_scale=False, formatter=None, legend='auto', capsize=0, err_kws=None, ci=<deprecated>, errcolor= <deprecated>, ax=None, **kwargs)

Function	Description	Parameters
sns.countplot()	Shows the count of observations in each categorical bin.	seaborn.countplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, orient=None, color=None, palette=None, saturation=0.75, fill=True, hue_norm=None, stat='count', width=0.8, dodge='auto', gap=0, log_scale=None, native_scale=False, formatter=None, legend='auto', ax=None, **kwargs)
sns.histplot()	Plots a histogram with optional KDE curve.	seaborn.histplot(data=None, *, x=None, y=None, hue=None, weights=None, stat='count', bins='auto', binwidth=None, binrange=None, cumulative=False, common_norm=True, multiple='layer', element='bars', shrink=1, kde=False, kde_kws=None, thresh=0, pmax=None, cbar=False, palette=None, hue_norm=None, color=None, log_scale=None, legend=True, ax=None, **kwargs)

Function	Description	Parameters
sns.kdeplot()	Plots the kernel density estimate of a distribution.	seaborn.kdeplot(data=None, *, x=None, y=None, hue=None, weights=None, palette=None, hue_order=None, hue_norm=None, color=None, fill=None, multiple='layer', common_norm=True, common_grid=False, cumulative=False, bw_method='scott', bw_adjust=1, warn_singular=True, log_scale=None, levels=10, thresh=0.05, gridsize=200, cut=3, clip=None, legend=True, cbar=False, cbar_ax=None, cbar_kws=None, ax=None, **kwargs)
sns.boxplot()	Displays a box-and-whisker plot.	seaborn.boxplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, orient=None, color=None, palette=None, saturation=0.75, width=0.8, gap=0, whis=1.5, linecolor='auto', fliersize=None, hue_norm=None, log_scale=None, formatter=None, legend='auto', ax=None, **kwargs)

Function	Description	Parameters
sns.violinplot()	Displays a violin plot combining KDE and boxplot.	seaborn.violinplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, orient=None, color=None, palette=None, saturation=0.75, fill=True, inner='box', split=False, width=0.8, dodge='auto', gap=0, linewidth=None, linecolor='auto', cut=2, gridsize=100, bw_method='scott', bw_adjust=1, density_norm='area', common_norm=False, bw=<deprecated>, inner_kws=None, ax=None, **kwargs)
sns.heatmap()	Visualizes matrix data using color encoding.	seaborn.heatmap(data, *, vmin=None, vmax=None, cmap=None, center=None, robust=False, annot=None, fmt='.2g', annot_kws=None, linewidths=0, linecolor='white', cbar=True, cbar_kws=None, cbar_ax=None, square=False, xticklabels='auto', mask=None, ax=None, **kwargs)

Function	Description	Parameters
sns.pairplot()	Plots pairwise relationships in a dataset.	seaborn.pairplot(data, *, hue=None, hue_order=None, palette=None, vars=None, x_vars=None, y_vars=None, kind='scatter', diag_kind='auto', markers=None, height=2.5, aspect=1, corner=False, dropna=False, plot_kws=None, diag_kws=None, grid_kws=None, size=None)
sns.displot()	Provides a figure-level distribution plot.	seaborn.displot(data=None, *, x=None, y=None, hue=None, row=None, col=None, weights=None, kind='hist', rug=False, rug_kws=None, log_scale=None, legend=True, palette=None, hue_order=None, hue_norm=None, color=None, col_wrap=None, row_order=None, col_order=None, height=5, aspect=1, facet_kws=None, **kwargs)

Function	Description	Parameters
sns.catplot()	Provides a categorical plot interface.	<pre>seaborn.catplot(data=None, *, x=None, y=None, hue=None, row=None, col=None, kind='strip', estimator='mean', errorbar=('ci', 95), n_boot=1000, seed=None, units=None, weights=None, order=None, color=None, palette=None, hue_norm=None, legend='auto', legend_out=True, sharex=True, sharey=True, margin_titles=False, facet_kws=None, ci=<deprecated>, **kwargs)</pre>

Function	Description	Parameters
sns.regplot()	Plots data and a linear regression model fit.	seaborn.regplot(data=None, *, x=None, y=None, x_estimator=None, x_ci='ci', scatter=True, fit_reg=True, ci=95, n_boot=1000, units=None, seed=None, order=1, logistic=False, lowess=False, robust=False, logx=False, x_partial=None, truncate=True, marker='o', scatter_kws=None, line_kws=None, ax=None)
sns.jointplot()	Draws a bivariate plot with marginal distributions.	seaborn.jointplot(data=None, *, x=None, y=None, hue=None, kind='scatter', height=6, ratio=5, space=0.2, dropna=False, xlim=None, ylim=None, color=None, palette=None, hue_order=None, hue_norm=None, marginal_ticks=False, joint_kws=None, marginal_kws=None, **kwargs)

Function	Description	Parameters
sns.stripplot()	Displays a scatter plot for categorical data.	seaborn.stripplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, jitter=True, dodge=False, orient=None, color=None, palette=None, size=5, edgecolor=<default>, linewidth=0, hue_norm=None, log_scale=None, native_scale=False, formatter=None, legend='auto', ax=None, **kwargs)
sns.swarmplot()	Displays categorical scatter plots with non-overlapping points.	seaborn.swarmplot(data=None, *, x=None, y=None, hue=None, order=None, hue_order=None, dodge=False, orient=None, color=None, palette=None, size=5, edgecolor=None, linewidth=0, hue_norm=None, log_scale=None, native_scale=False, formatter=None, legend='auto', warn_thresh=0.05, ax=None, **kwargs)

Function	Description	Parameters
sns.ecdfplot()	Plots the empirical cumulative distribution function.	seaborn.ecdfplot(data=None, *, x=None, y=None, hue=None, weights=None, stat='proportion', complementary=False, palette=None, hue_order=None, hue_norm=None, log_scale=None, legend=True, ax=None, **kwargs)
sns.lmplot()	Plots regression models across subsets of data.	seaborn.lmplot(data, *, x=None, y=None, hue=None, col=None, row=None, palette=None, col_wrap=None, height=5, hue_order=None, col_order=None, x_estimator=None, x_bins=None, x_ci='ci', scatter=True, fit_reg=True, ci=95, n_boot=1000, units=None, seed=None, order=1, logistic=False, lowess=False, robust=False, logx=False, x_partial=None, y_partial=None, truncate=True, x_jitter=None, y_jitter=None, scatter_kws=None, line_kws=None, facet_kws=None)

Function	Description	Parameters
sns.color_palette()	Sets and retrieves color palettes.	seaborn. color_palette(palette=None, n_colors=None, desat=None, as_cmap=False)
sns.set_style()	Sets the visual theme of seaborn plots.	seaborn.set_style(style=None, rc=None)
sns.despine()	Removes the top and right spines from plots.	seaborn.despine(fig=None, ax=None, top=True, right=True, left=False, bottom=False, offset=None, trim=False)
sns.load_dataset()	Loads sample datasets provided by seaborn.	seaborn. load_dataset (name, cache=True, data_home=None, **kws)

For details on operations in seaborn, see the official documentation: <https://seaborn.pydata.org/tutorial/introduction.html>